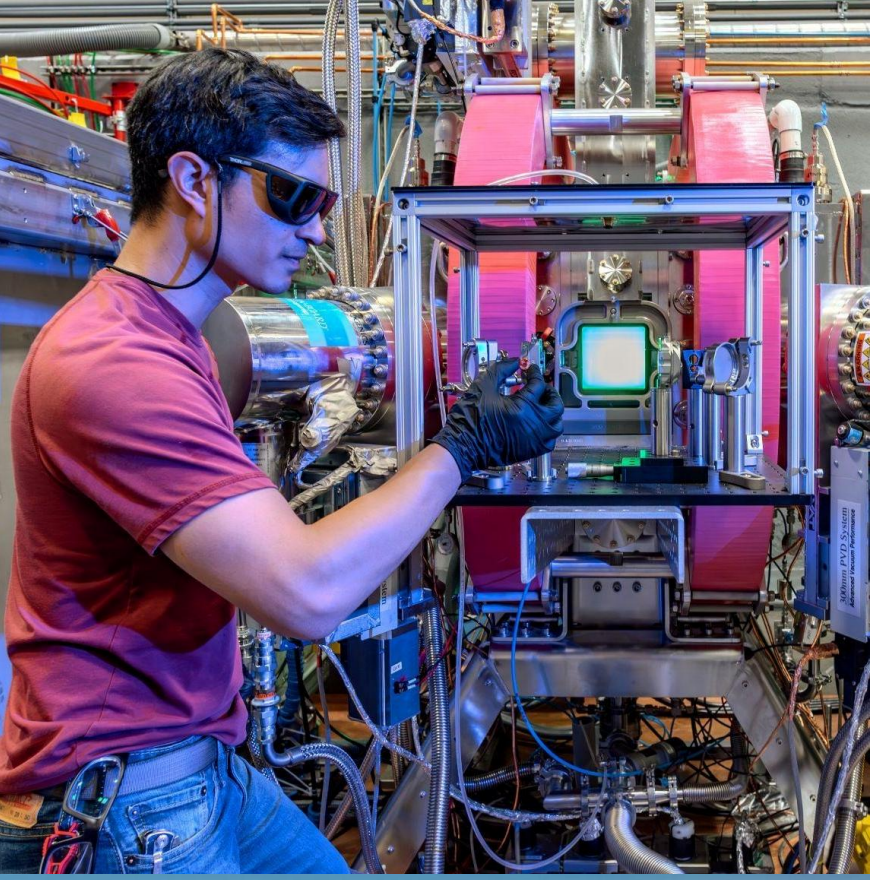




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HYbrid Pumped ExcimeR Laser (HYPER-LASER)

Jake Martin, Summer Andrews, Steven Pugh, Brian Drews, Chase Hamilton, Em Bulkowski, Dr. Dmitrii Andreev, Dr. Austin Steinforth, Dr. Gary Eden (UIUC), Dr. Zhenhuan Yi (TAMU), Dr. Matt Wolford (NRL), Dr. Andrey Mironov, Dr. Kevin Finch

Award No. DE-AR0002030

Project Kickoff and Reporting Quarter 1

March 13, 2026

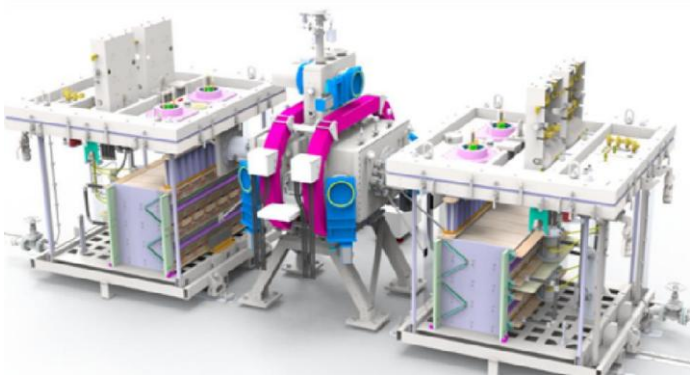
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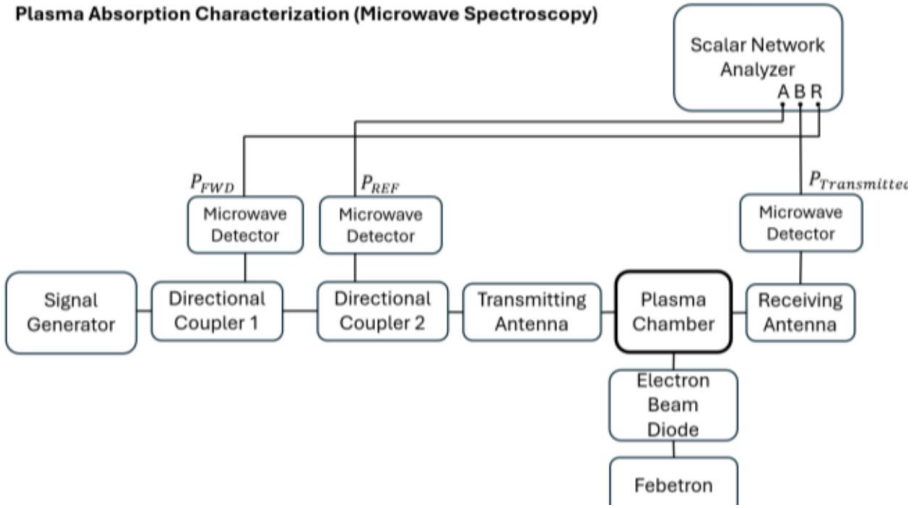
HYbrid Pumped ExcimeR Laser (HYPER-LASER) - One Slide Summary

KrF Laser Architecture

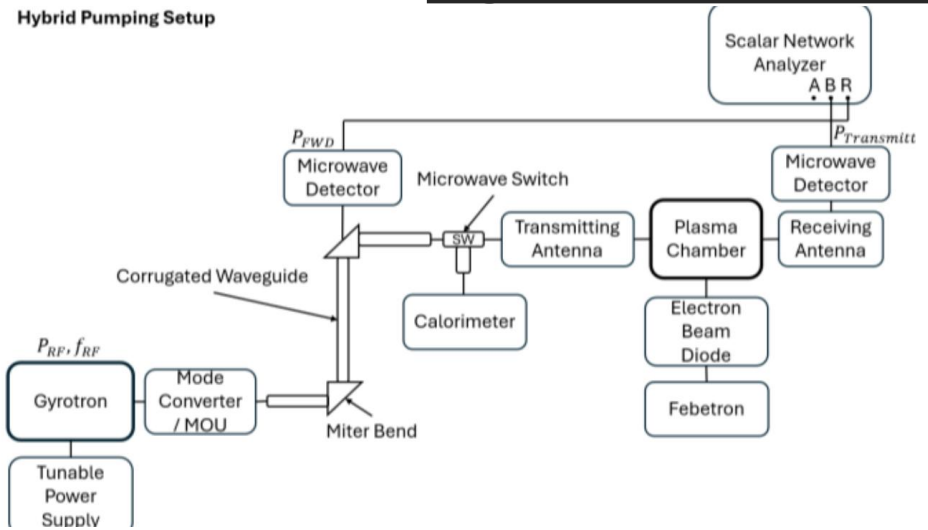
~6 ft Tall

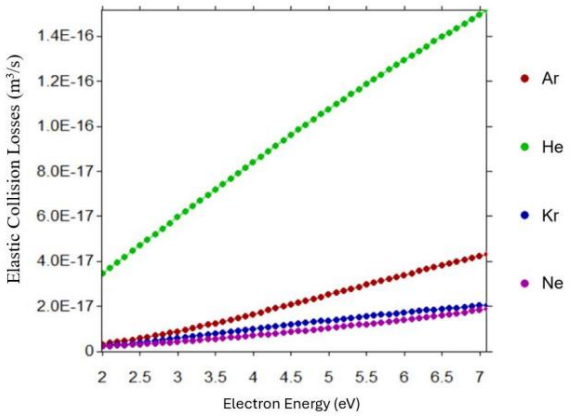


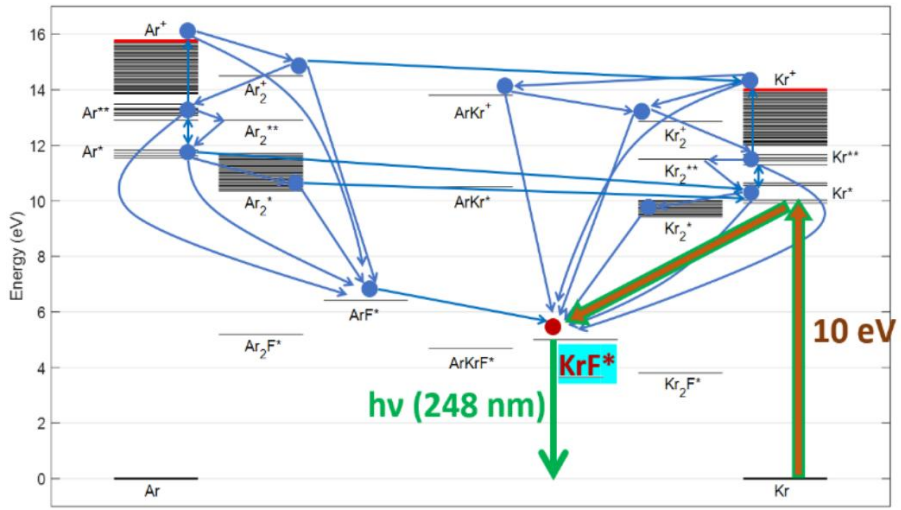
Low-Power Preliminary Study

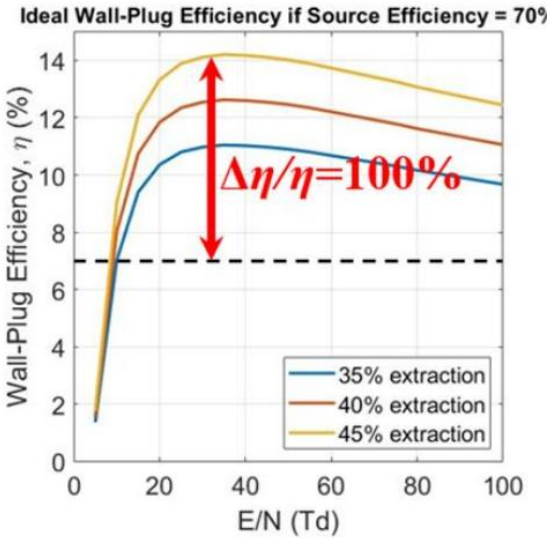


High-Power HYbrid Pumping









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Team



Zhenhuan Yi, PhD
Co-PI/TAMU Lead
Laser-Plasma Kinetic Simulations
Plasma/Electromagnetic Interactions



Kevin Finch, PhD
PI/Technical Lead
SBIR Contracts, Instrumentation Development, Plasma Diagnostics, Experiment and System Design



Matt Wolford, PhD
Co-PI/NRL Lead
Head Electron Beam Science & Applications Section - Laser Plasma Branch - Plasma Physics Division



Andrey Mironov, PhD
Director of Laser Experimental Science
High-Power Lasers/Spectroscopy, Experiment and System Design



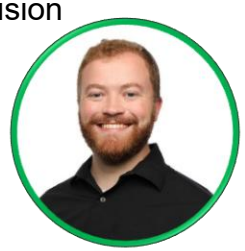
J. Gary Eden, PhD
Co-PI/UIUC Lead
High-Power KrF Laser Design, Plasma/Microwave Interactions



Emily (Em) Bulkowski
Systems Engineer
Project and System Objective Alignment



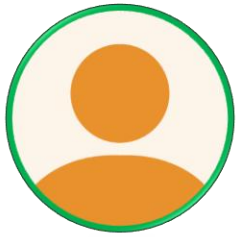
Brian Drews
Senior Manager
System Integration, Gas/Vacuum Systems



Austin Steinforth, PhD
Chief Laser Architect
System Integration, KrF Laser Design, Plasma Kinetics



Jacob (Jake) Martin
Pulsed Power Physicist and Computational Analyst
E-Beam and Microwave Delivery, Data Analysis



Chase Hamilton
Systems Engineer
Project and System Objective Alignment



Steven (Steve) Pugh
Sr. Mechanical Design Engineer/KJC Element Lead
System Mechanical Design



Dmitrii Andreev, PhD
Sr. Pulsed Power Physicist/Engineer
E-Beam and Microwave Delivery, Design



Summer Andrews
Pulsed Power Design Engineer
E-Beam and Microwave Delivery, Simulations

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Internal Meetings – Weekly
External Meetings - Monthly

Collaborators:

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U.S. NAVAL RESEARCH LABORATORY

Executive Summary

Total funds awarded/spent:
$4,829,466 / $100,447
POP end date: 11/17/28

Technology Summary

Goal: Assess feasibility of a new hybrid-pumping approach to improve the intrinsic efficiency of krypton-fluoride (KrF) deep-UV lasers.

Concept: Combine electron-beam pumping with microwave excitation to boost KrF laser wall-plug efficiency beyond conventional e-beam methods.

Approach: Design and build a small-scale experimental setup to optimize the concept and validate associated simulations.

Technology Progress

- *Project Team Established and Budget approved.*
- *New Project Technical Lead and Principal Investigator*
- *Subawards issued to TAMU and UIUC for negotiations*
- *M1.1 – Project Schedule-of-Work (GANTT) Finalized.*
- *M2.1 – Febetron (E-Beam Power Supply) Acquired.*
- *M2.2 – Initial System Design/Facility Needs Underway*
- *M2.2 – P&ID, ConOps, ADD, and MIPs in Progress*

T2M

- *Clear commercialization path via Xcimer's IFE facility roadmap from prototype(s) to grid power.*

Proposed Targets

Metric	State of the Art	Current	Proposed/Achievable
KrF Laser Wall-Plug Efficiency	~7%	~7%	>14%
Required IFE Target Gain	150-200	150-200	75-100

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Project Overview (Kickoff): M1.1 Refine Tasks/Milestones

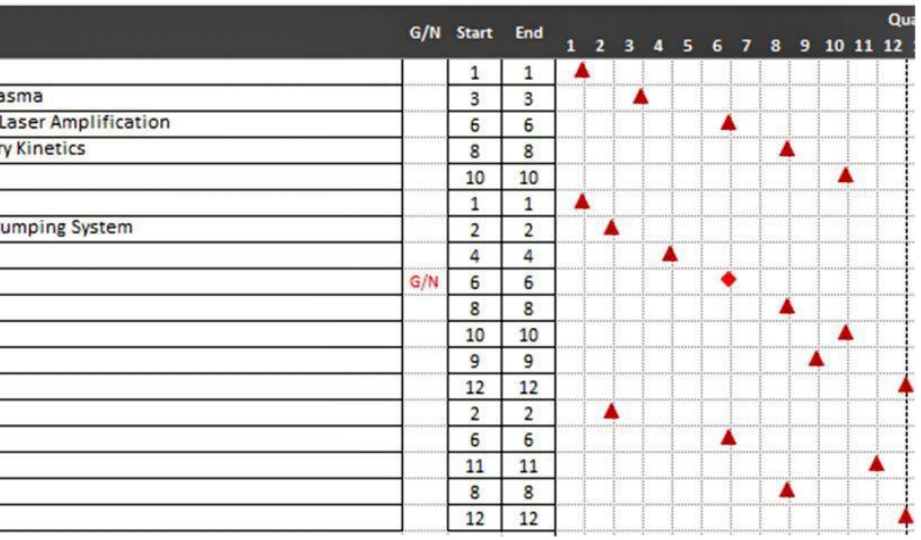
End: 05/18/26

- **Refine current Schedule-of-Work**
  - Define start and duration for each milestone
    - Need to run concurrently for parallelization

- **Evaluate Time/Budget to Possibly Include Following**
  - Include electron-heating sources over a broader frequency range
    - Microwave sources >100 GHz
      - mm-wave and THz sources.
  - Expand electron sources to include different e-beams designs
    - Cold cathodes (velvet, carbon, etc.)
    - Array (patterned) cathodes
      - Strips
      - Shaped Field Emitters

Original Submitted GANTT Chart

WBS	Task/Milestone Title	G/N	Start	End
M1.1	Refine tasks and milestones		1	1
M1.2	Simulation: Electromagnetic Wave Penetration into Plasma		3	3
M1.3	Kinetics Code Development for Plasma Chemistry and Laser Amplification		6	6
M1.4	Code Validations: Boltzmann-PIC and Plasma Chemistry Kinetics		8	8
M1.5	HYPER-Laser Optimizer		10	10
M2.1	Equipment procurement		1	1
M2.2	Design Experimental Setup, Including Electron Beam Pumping System		2	2
M2.3	Experimental Setup Built and Tested		4	4
M2.4	Microwave Absorption Measurements G/N	G/N	6	6
M2.5	Experimental Setup Upgraded and Tested		8	8
M2.6	Hybrid-Pumping Measurements		10	10
M3.1	Microwave Source Evaluation		9	9
M3.2	Commercial Feasibility of HYPER-Lasers		12	12
M5.1	Initial Technology-to-Market (T2M) plan		2	2
M5.2	Revised T2M Plan		6	6
M5.3	Final T2M Plan		11	11
M6.1	Publication		8	8
M6.2	Publication		12	12



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Project Overview (Kickoff): M1.2 Simulation: Electromagnetic Wave Penetration into Plasma

End: 11/18/26

- **Develop a hybrid Boltzmann–PIC simulation code**
  - Combine key computational methods
    - Particle-in-Cell (PIC) technique for electromagnetic field transport
    - Monte Carlo model to capture collisional interactions
  - Microwave, mm-wave, and THz field penetration, reflection, and propagation in excimer laser plasmas
  - Include plasma heating processes
  - Include collisional processes

- **Solve for various laser plasma conditions**
  - Electron number densities: 10^{12} -10^{15} cm^{-3}
  - Neutral gas pressures of ~0.5 - 3 atm
  - Various gas mixtures of Ar, Kr, He, Ne, F_2, NF_3

- **Model validation and fidelity assessment**
  - Plasma-wave interactions, ionization dynamics, and collisional processes occurring in <1 ps timescales
  - Temporal steady-state verification
  - Minimization of residuals within defined tolerance

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Project Overview (Kickoff): M1.3 Kinetics Code Development for Plasma Chemistry and Laser Amplification

End: 08/18/27

- **Develop a time-dependent plasma kinetics code to model reaction networks in the laser plasma**
  - Solve for coupled chemical rate equations
  - Incorporate species production rates (e.g., Kr, Kr^+, Kr*, KrF*, etc.)
    - Obtained directly from the Boltzmann solver
  - Include a comprehensive set of chemical reactions
    - Neutrals
    - Ions
    - Excited moieties

- **Compute time-dependent concentrations of all species at >1 ps timescales**
  - Model primary processes
    - 248.5 nm Fluorescence from various gas mixtures (Kr, Ar, F_2, He, Ne, NF_3)
    - Spatiotemporal profiles of energy deposition from excitation sources with <10 ns resolution.
  - Model additional processes
    - Absorption by the laser gas medium
    - Amplified spontaneous emission (ASE) in the 3D gas volume
    - Absorption at chamber walls

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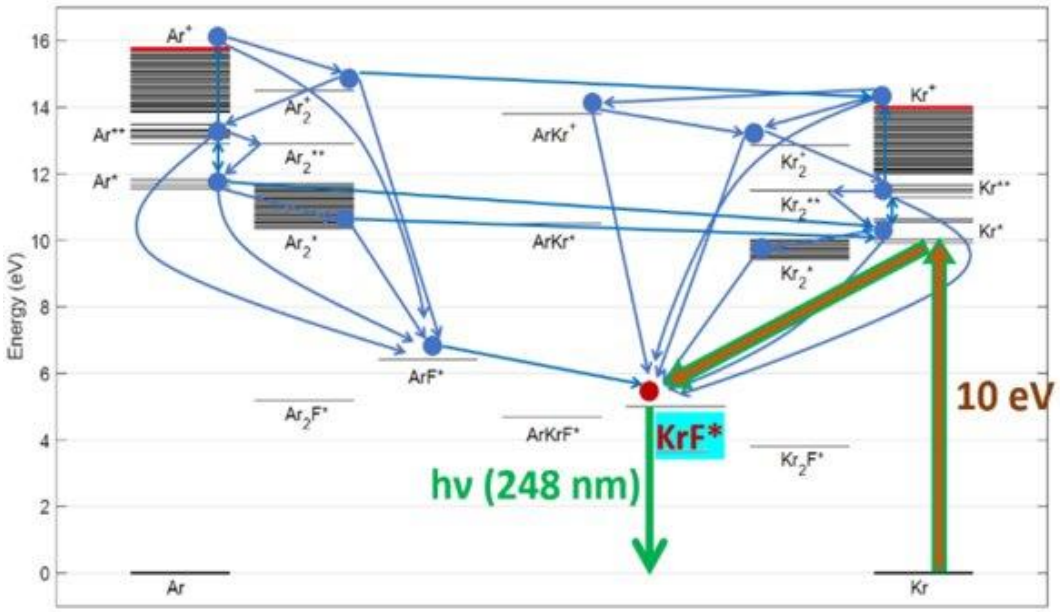
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Project Overview (Kickoff): M1.4 Code Validations: Boltzmann-PIC and Plasma Chemistry Kinetics

End: 02/18/28

- **Validate the Boltzmann Solver**
  - Electromagnetic propagation in gas/plasma
    - Use S11 and S21 measurements from task M2.4
  - Electromagnetic power deposition into the gas cell,
    - Include ionization and sustained discharge behavior
  - Obtain a target agreement of within 10%
    - Experiments vs Simulations



- **Validate the Plasma Kinetics Code**
  - Excimer chemistry reaction pathways
    - Optical emission spectroscopy (OES) of excited moieties
      - Spectra in the ~215 nm-500 nm wavelength range
  - Operation across gas pressures ~0.5 - 3 atm
  - Gas mixtures including Ar, Kr, He, Ne, F_2, NF_3
  - Obtain target agreement within 15% between simulations and experiments
    - Use results from task M2.6

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Project Overview (Kickoff): M1.5 HYPER-Laser Optimizer

End: 08/18/28

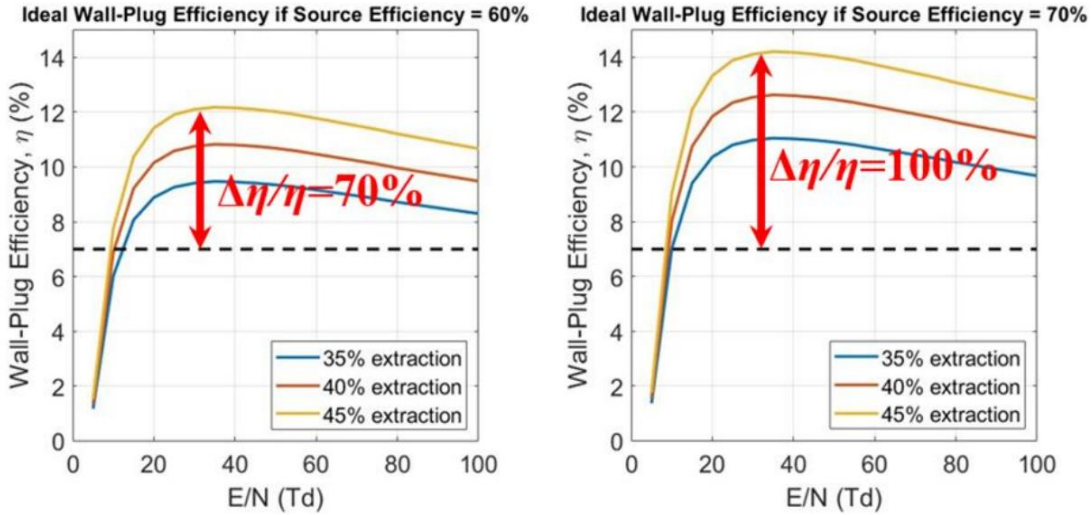
- **Merge Boltzmann-PIC and Plasma Kinetics Codes for Comprehensive Model of Hybrid Excitation**
  - Include all major physical and chemical processes across wide timescale
    - <1 ps using the Boltzmann-PIC Solver
    - >1 ps using the Plasma Kinetics Code

- **Predict KrF Lasing performance**
  - Use varying gas mixtures and excitation modalities
  - Determine system limits and operational boundaries
  - Find local optimizations that improve the laser's intrinsic efficiency
    - Needs to be greater than currently achieved value of ~7%

$$\eta_{extract}^{max} = \left(1 - \sqrt{\frac{\alpha}{g_0}}\right)^2$$

g_0/α	$\eta_{extract}^{max}$
6	35%
8	42%
10	47%
12	51%
15	55%

- **Test and Validate the HYPER-Laser Optimizer**
  - Operation across gas pressures ~0.5 - 3 atm
  - Gas mixtures including Ar, Kr, He, Ne, F_2, NF_3
  - Obtain target agreement within 15%
    - Between simulations and experiments
    - Use results from task M2.6



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Project Overview (Kickoff): M2.1 Equipment Procurement

End: 05/18/26

- **High-Power Microwave Sources**
  - GHz pulsed, >100 kW power, waveguide-coupled output, TE12,2 mode
- **Low-Power Microwave Measurement Hardware**
  - Components to measure S11 and S21 during plasma interaction
    - Network Analyzer, Waveguides, Miter Bends, Directional Couplers, Load
  - 10-26 GHz range
  - 26-100 GHz range
  - ~20 mW power
- **Laser/Plasma Chamber and Associated Hardware**
  - ~5 cm to ~10 cm Pumped Volume Capable
  - ≥12 ports
  - Interfaces to microwave, E-beam, and diagnostic suites
- **Plasma Diagnostic Equipment**
  - Optic Table
  - iCCD Camera/Monochromator
  - Collection Optics
- **Vacuum Equipment (Diode & Laser/Plasma Chamber)**
  - Roughing pump(s)
  - Turbo pump(s) capable of reaching 10^{-6} Torr
  - Pressure Transducer(s)
  - Valves, connectors, lines
- **Gas Mixing/Delivery**
  - Mass flow controller(s)
  - Valves, connectors, lines
- **Timing**
  - Digital delay generator (DDG)
  - SMA/BNC cables and connectors
- **Other Diagnostics**
  - Fast Photodiode(s)
  - Power meter(s)
  - Oscilloscope(s)

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Project Overview (Kickoff): M2.2 Design Experimental Setup, Including E-Beam System

End: 08/18/26

- **Develop a detailed schematic of the full experimental setup (chambers, layout, equipment, diagnostics, inputs/outputs, operational limits, data handling).**
  - E-beam Diode reaches <10^{-6} Torr.
  - Design an electron gun using Febetron
    - Pulse forming network
  - Laser/Plasma chamber capable of
    - Electron number densities: 10^{12} -10^{15} cm^{-3}
    - Neutral gas pressures of ~0.5 - 3 atm
    - Various gas mixtures of Ar, Kr, He, Ne, F_2, NF_3



Febetron 706

  - Setup supports variable plasma aspect ratios and tunable microwave sources (~10–100 GHz).
  - Plasma geometry depends on cold-cathode and hibachi transmission-window design.
  - Diagnostics include photodiodes, oscilloscopes, microwave detectors, optical emission spectroscopy, laser absorption/scattering and electrical probes.
  - System designed for flexibility and post-measurement modifications (after Milestone M2.4).

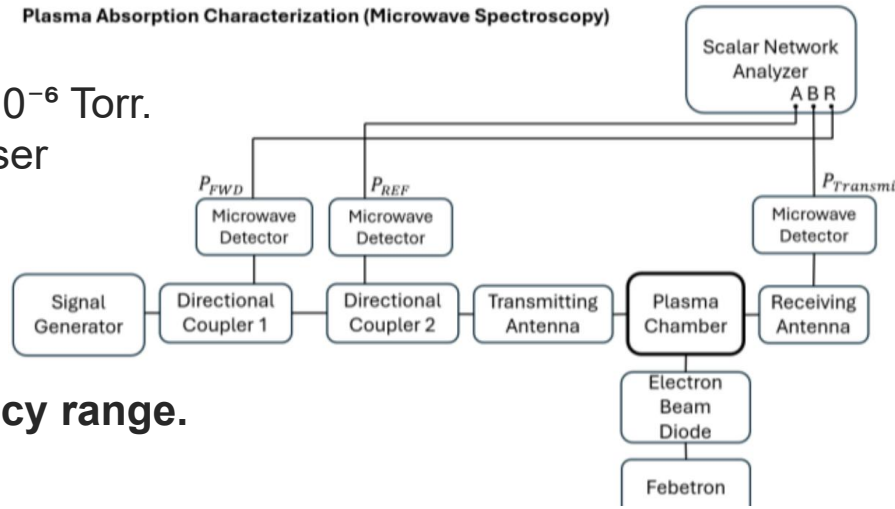
- **Present finalized plans to the Program Director for approval.**

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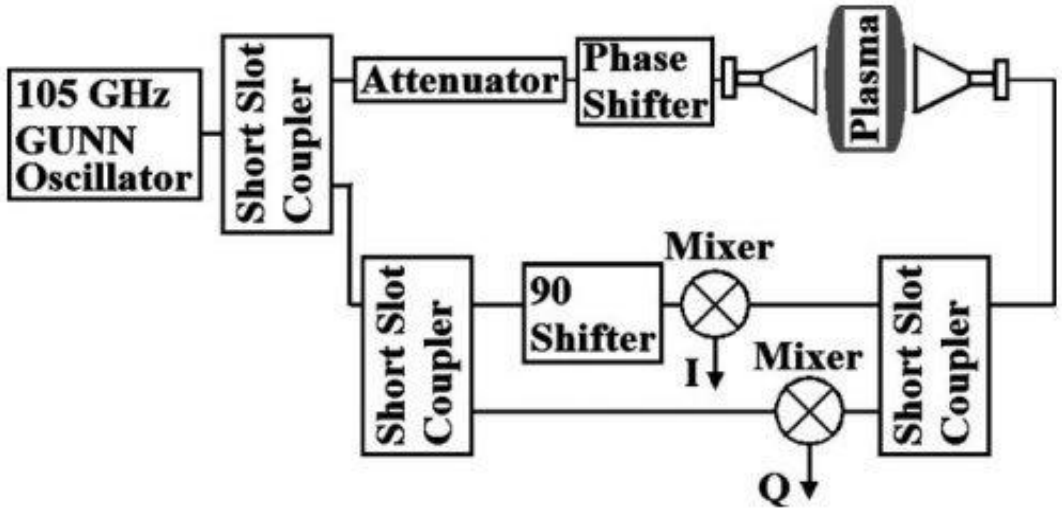
Project Overview (Kickoff): M2.3 Experimental Setup Built and Tested

End: 02/18/27

- **Construct the experimental setup defined in M2.2 to study microwave heating of electrons in excimer-laser plasmas**
  - Achieve electron densities of 10^{12}–10^{15} cm^{-3}.
    - Adjust the plasma aspect ratio across the range defined in M2.2.
  - Verify that the main vacuum chamber (diode) achieves pressures better than 10^{-6} Torr.
  - Generate an electron beam using a cold cathode connected to a Febetron pulser
    - Confirm pulse parameters match those specified in M2.2
    - Identify the need for Magnetic Field?
      - To prevent electron beam focusing/diverging effects



- **Test the tunable microwave source from M2.1 across its full specified frequency range.**
  - Variable frequency ~10-100 GHz
  - Scattering Parameter Measurement Testing
    - S11 (reflection) and S21 (transmission)
    - Measure spatiotemporal resolution
    - Check wave mode in the plasma interaction regior



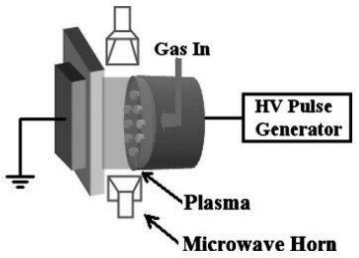
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Lu, P., X., and Laroussi, M., Applied Physics Letters 92, 051501 (2008)

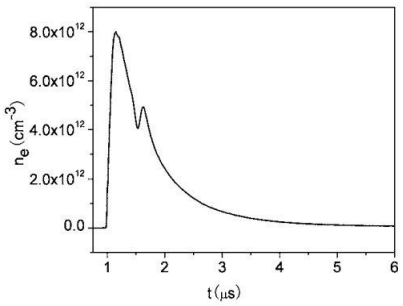
Project Overview (Kickoff): M2.4 Microwave Absorption Measurements (Go/No-Go)

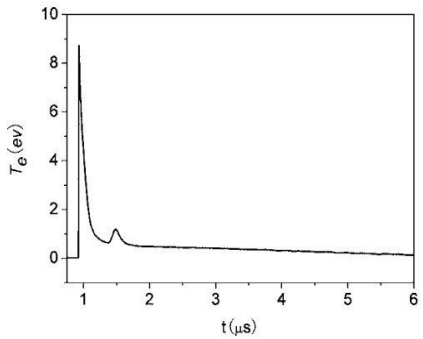
End: 08/18/27

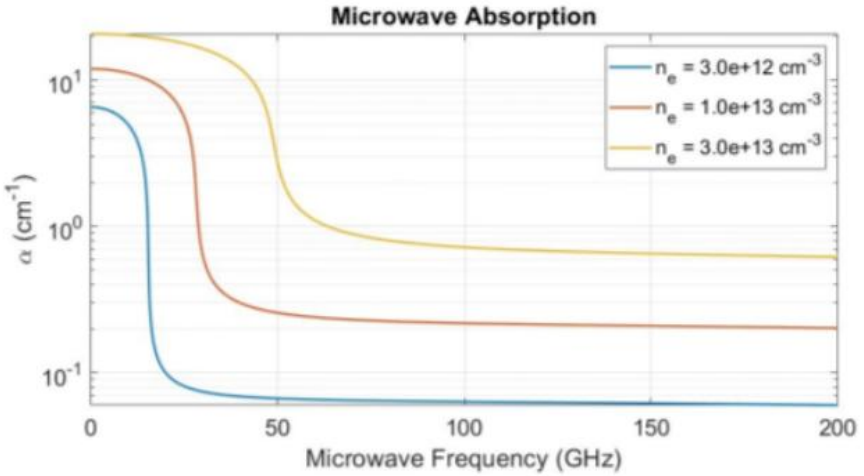
- **Use low-power (mW) microwave sources to measure S21 in electron-beam pumped KrF plasmas.**
  - Conduct experiments across a range of:
    - Gas mixtures: Kr (0–40%), Ar (60–100%), F_2 (0–2%)
    - Pressures: 0.5–3 atm
    - Plasma dimensions defined by
      - Electron densities: 10^{12}–10^{15} cm^{-3}
      - Hibachi transmission-window and cathode designs from M2.2
- **Confirm measured absorption at all frequencies is consistent with M1.2 simulation results.**
  - Use preliminary 0D calculations as a reference
  - For $n_e = 10^{13}$ cm^{-3} at 100 GHz, expected absorption coefficient ≈ 0.2 cm^{-1}.
  - Verify measured absorption at all frequencies from ~10-100 GHz
    - Final tolerance to be refined in M1.2
- **Microwave Interferometry**
  - Correlate phase/amplitude shifts to plasma collisional frequency and electron number density
  - Will estimate electron temperature under certain conditions

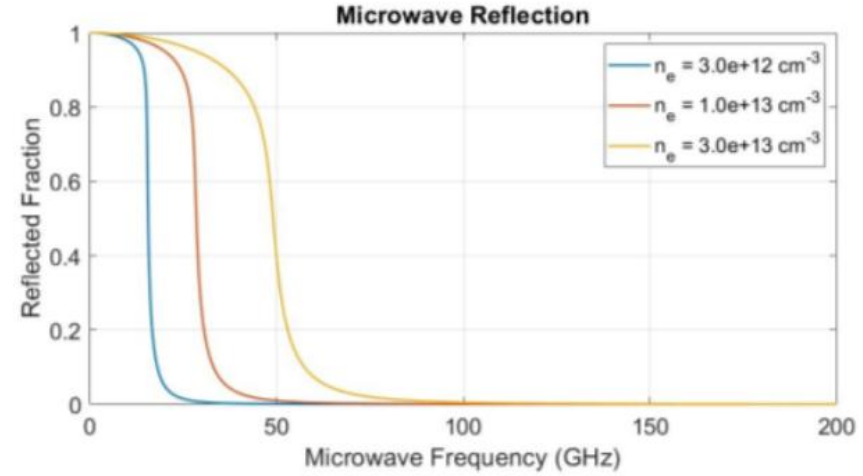


Lu, P., X., and Laroussi, M., Applied Physics Letters 92, 051501 (2008)









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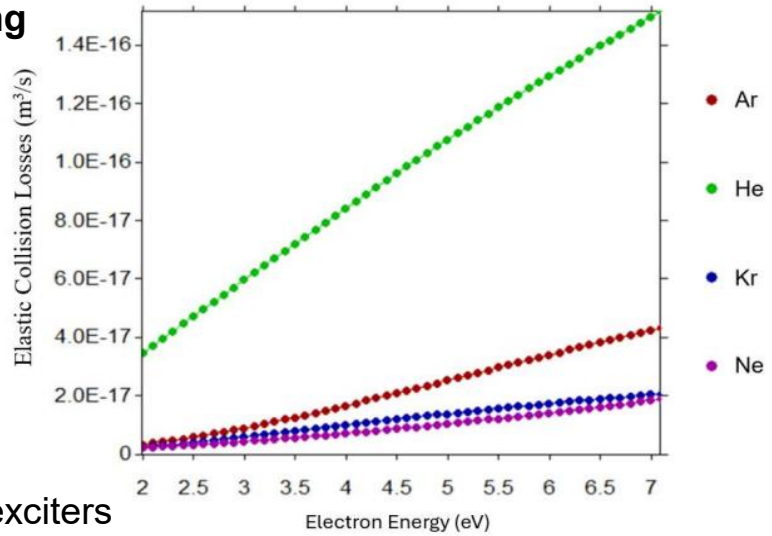
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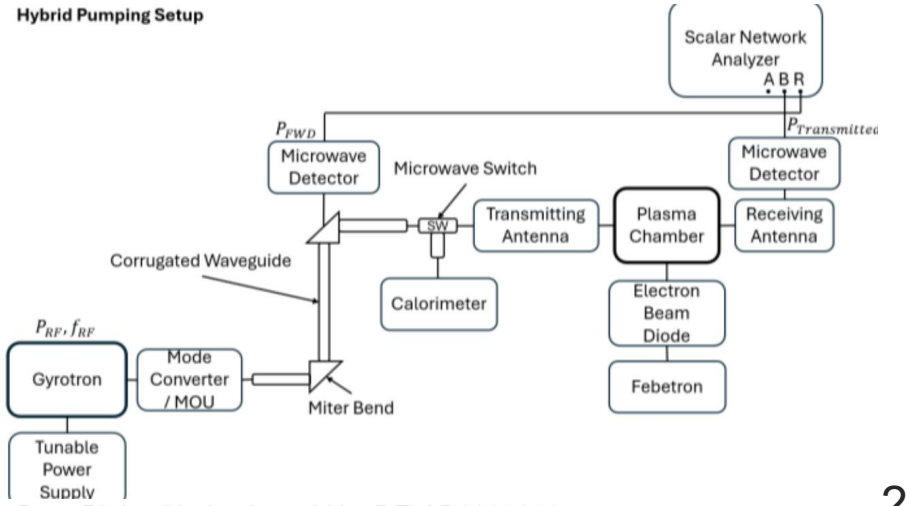
Project Overview (Kickoff): M2.5 Experimental Setup Upgraded and Tested

End: 02/18/28

- **Design, build, and commission a hybrid-pumping system combining**
  - A Febetron-based electron gun
  - A high-power pulsed microwave source
    - Will refine specs during project
      - Preliminary: >0.1 MW output, $TE_{12,2}$ mode
  - Spectrometers and monochromators
  - Photodiodes and oscilloscopes
  - Microwave sensors
  - Electrical probes for current and voltage measurements



- **Excite excimer laser gas mixtures using hybrid excitation**
  - Variable pulse widths and relative timing delays between the two exciters
  - Use optical emission spectroscopy (OES) Plasma Diagnostics
    - Analyze gas-phase kinetics under varying microwave and plasma conditions.





2 MW Gyrotron (95 GHz)

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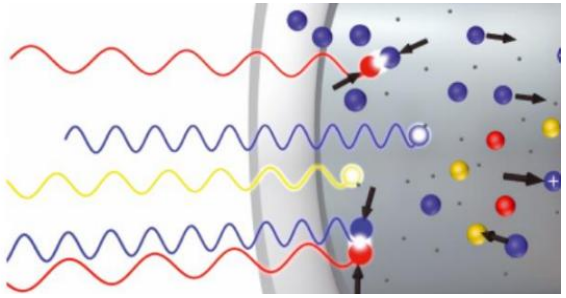
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Project Overview (Kickoff): M2.6 Hybrid Pumping Measurements

End: 08/18/28

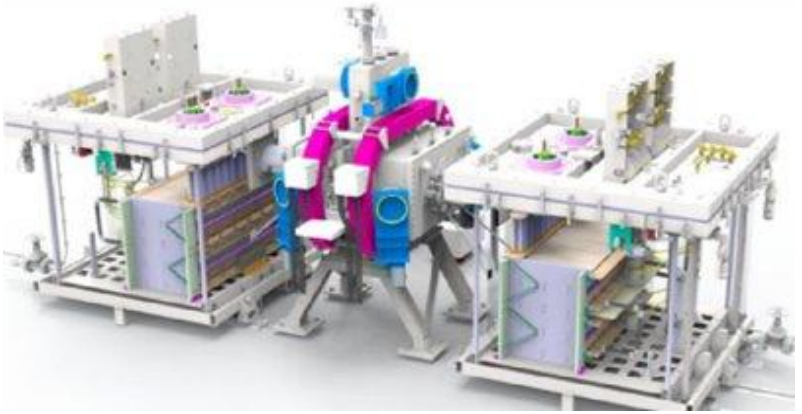
- **Optimize high-power microwave sources with electron beam and measure KrF* emission during hybrid excitation**
  - Measure sidelight 248.5 nm fluorescence measurements along microwave beam path
    - Relate this to achievable laser gain through direct proportionality
- **Start with testing local optimizations found in task M1.5**
  - Vary the temporal overlap of the electron beam and the microwave pulses
    - Gas Kinetic Changes
    - Absorption Effects
- **Optimize hybrid-pumping performance by measuring:**
  - Fluorescence intensity of the KrF* excimer at 248.5 nm
    - Directly proportional to the small-signal gain at the same wavelength.
- **Calculate the increase in fluorescence efficiency and corresponding wall-plug efficiency improvement from adding microwave excitation**
  - Use models developed in M1.3 and M1.4.

Final target performance metrics will be reviewed and approved by the Program Director.



Adapted From: https://www.spectruma.de/en/resources/gdoes-theory

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Project Overview (Kickoff): M3.1 Microwave Source Evaluation

End: 05/18/28

- **Survey commercially available microwave sources relevant to high-power laser-driving applications and estimate cost/availability.**
  - Physics/fundamental science testing
  - Design of new microwave sources
  - Construction of systems capable of powering an LPK-scale laser (≈10 cm × 10 cm × 50 cm).
  - Use initial parameters (power, frequency, system requirements) informed by simulations from M1.4.
  - Refine system specifications based on experimental results from M2.6.
  - Re-evaluate and update requirements as needed throughout the study.

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Project Overview (Kickoff): M3.2 Commercial Feasibility of HYPER-Lasers

End: 11/17/28

- **Assess the commercial viability of a HYbrid-Pumped ExcimeR Laser (HYPER-Laser) compared to a purely electron-beam pumped system.**
  - Using results from M2.5, analyze key trade-offs between the two approaches, including:
    - Capital cost
    - Operating cost
    - System complexity
    - System size
    - Generator Efficiency
    - Maintenance/Down Time

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Project Overview (Kickoff): M5.1, M5.2, M5.3 – Initial, Revised, Final T2M Plan

- **Consult with the Program Director and T2M Advisor to submit the first draft of the T2M plan.**
  - Ensure the plan addresses, throughout the project:
    - IP freedom to operate
    - IP protection for project-generated technologies
    - Techno-economic analysis
    - Customer engagement and potential partnerships
    - Supply-chain considerations
    - Adoption risks
    - Follow-on funding opportunities

Initial: **08/18/26**

Revised: **08/18/27**

Final: **08/18/28**

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Project Overview (Kickoff): M6.1, M6.2 – Publications

- **Submit at least one manuscript for publication in an appropriate peer-reviewed journal**
  - Content based on results from experimentation and simulations
  - Will publish non-enabling results and data
    - Consistent with "Protected Data Provisions"

Manuscript 1 **02/18/28**
Manuscript 2 **11/17/28**

- Will share data necessary to validate published research findings and reported performance metrics, consistent with DOE/ARPA-E requirements and subject to protection of all proprietary and enabling details.

- Shared data will be sufficient to substantiate reported conclusions, but will exclude proprietary process parameters, detailed fabrication protocols, source code, trade-secret information, and any other enabling details.

- Data will be shared through publications consistent with the contract deliverables.

Quarter 1 Updates - Summary

- **Team Assembled and Budget Approved**
  - New Project Technical Lead/Principal Investigator
- **Changed SoW to reflect agreed upon 3-month milestone shift**
  - Revised milestones to have concurrent timelines
- **Milestone Progress**
  - M1.1 – Project Schedule-of-Work (GANTT) Finalized.
  - M2.1 – Febetron (E-Beam Power Supply) Acquired.
  - M2.2 – Initial System Design/Facility Needs Underway
    - Block Diagrams
    - Plumbing and Instrumentation Diagram (P&ID)
    - Concept of Operations (ConOPs)
    - Architecture description design (ADD)
    - Most important requirements (MIPs)
  - M2.2 – Plasma Pumped Volume Determination
    - Preliminary E-beam/Plasma Kinetic simulations

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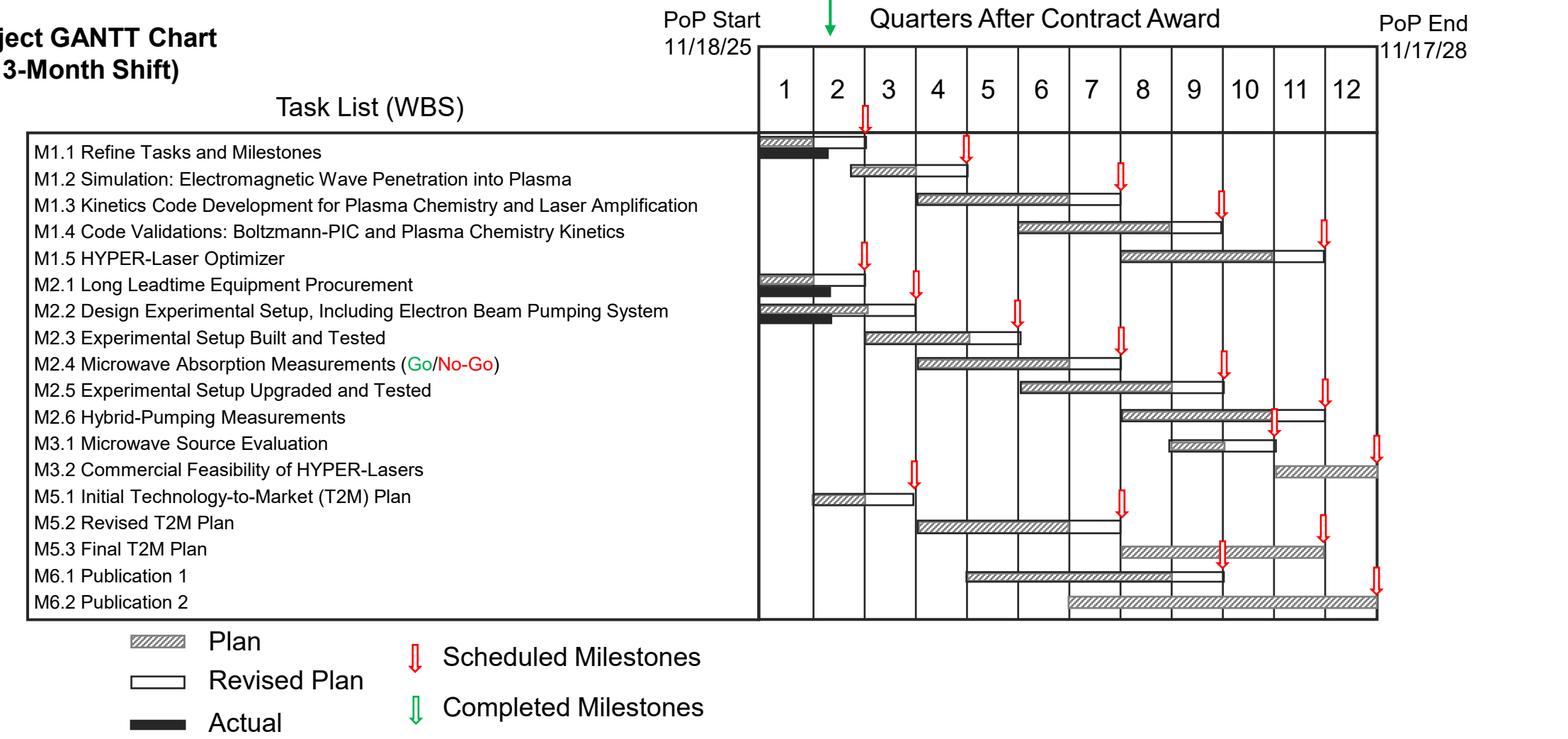
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Quarter 1 Updates: M1.1 Refine Tasks/Milestones - Updated GANTT

New Project GANTT Chart (With 3-Month Shift)

We Are Here 03/17/26

Begin: 11/18/25 **End: 05/18/26**



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Quarter 1 Updates: M2.1 Long Leadtime Equipment Procurement – Febetron (Cost Share)

Begin: 11/18/25 **End: 05/18/26**

Pulsed Power for Electron Beam (Febetron 701/266)



Output Voltage (Matched Load) – 600 kV
Output Current (Matched Load) – 6 kA
Output Impedance – 100 Ohms

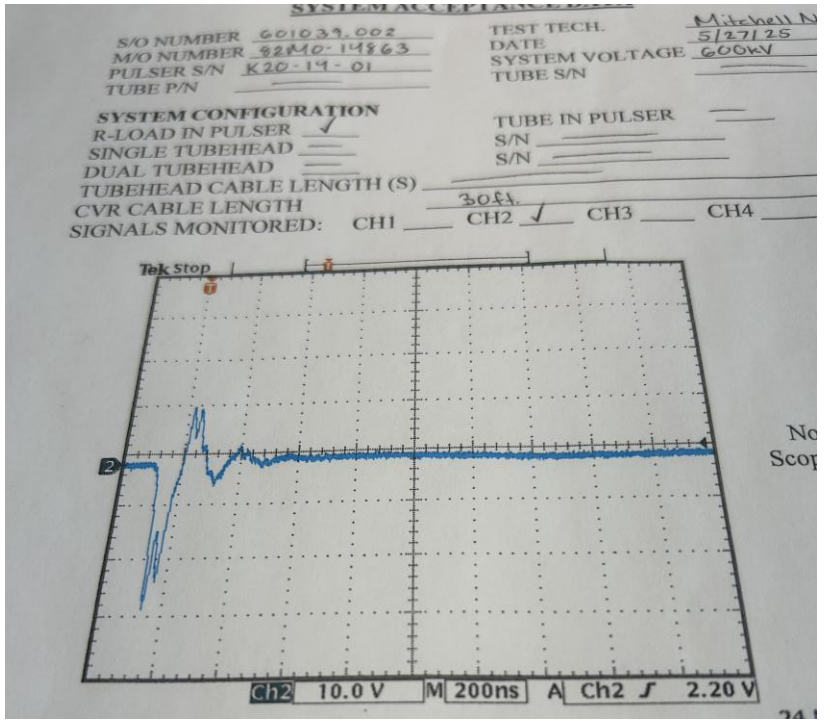
Pulse Generator (Trigger)



Cabling for Power Delivery



~50-75 ns (FWHM) Tested Pulse



Need to purchase a high (+/-35 kV) power supply.

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Quarter 1 Updates: M2.1 Long Leadtime Equipment Procurement – Low Power Microwave Equipment

Begin: 11/18/25 **End: 05/18/26**

Low Frequency system

- 1-26.5 GHz swept frequency source
- HP8756 Scalar network analyzer
  - 2 channels and a reference for normalization
- Waveguide directional couplers
- Waveguide detectors
- WR90 8-12 GHz
- WR62 12-18 GHz
- WR42 18-26 GHz



High Frequency Micro-Now Generator

- 705C
- Provides 27.7 KHz modulation for scalar network analyzer 8756 & 8757
- Swept or fixed frequency operation
- BWO heads
  - 26-40 GHz. WR28
  - 33-50 GHz WR19
  - 50-75 GHz WR-15
  - 75-110 GHz WR-10
  - 110-170 GHz WR-6
- Also HP8610 system for 26-40 GHz
- Waveguide components for most bands



Waveguide test equipment components







33-50 GHz 50-75 GHz



75-110 GHz



110-170 GHz



Tim Bigelow – MRP Technologies, Inc.

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Quarter 1 Updates: M2.2 Design Experimental Setup - Systems Engineering

Begin: 11/18/25 **End: 08/18/26**

- **Concept of Operations (ConOps)**
  - Outline created to compile information
  - Meeting with SMEs to describe how system will be operated and maintained
  - Conversation focuses on needed capabilities, functionality, safety, security, etc.
  - Intent is to derive the system needs to incorporate while still in early design
    - Access restrictions → access control system
    - Data collection → diagnostic selection, access control, & data visualization method

1. Purpose
2. Scope
3. Ground Rules and Assumptions
4. Operational Modes
5. Shot Operations
   5.1 Experiment Campaign Planning
   5.# Experiment Day
   5.#. Post Experiment Review
6. Maintenance Operations
7. Additional Operational Considerations
   7.#. Access Restrictions
   7.#. Data Collection
   7.#. Facility Needs
   7.#. Control Room
   7.#. Interlock System
8. Derived Needs from ConOps
9. References
10. Abbreviations and Definitions

- **Most Important Requirements (MIR)**
  - Drafting top level requirements to define minimum functionality to achieve objectives
  - Plan to decompose these to the sub-systems and component levels

- **Interface Block Diagram (IBD)**
  - Maps the physical and logical interfaces between sub-systems
  - Enables visualization and understanding of how sub-systems will interact

Interface ID	Interface Type	From-To
LI-1	Laser	Laser - Reaction Chamber
EI-1	E-Beam	E-Beam - Reaction Chamber
MI-1	Microwave Energy	Microwave Source <-> Reaction Chamber
MI-2	Microwave Energy	Chamber <-> Microwave Sink
DI-1	Data	C&D <-> Reaction Chamber
DI-2	Data	C&D <-> Laser
DI-3	Data	C&D <-> E-Beam
DI-4	Data	C&D <-> Microwave Source
DI-5	Data	C&D <-> Microwave Sink
SI-1	Gas, Power	Site <-> Reaction Chamber
SI-2	Power	Site <-> Laser
SI-3	Power, Chilled Water	Site <-> E-Beam
SI-4	Power	Site <-> Microwave Source
SI-5	Power, Chilled Water	Site <-> Microwave Sink
SI-6	Power	Site <-> C&D

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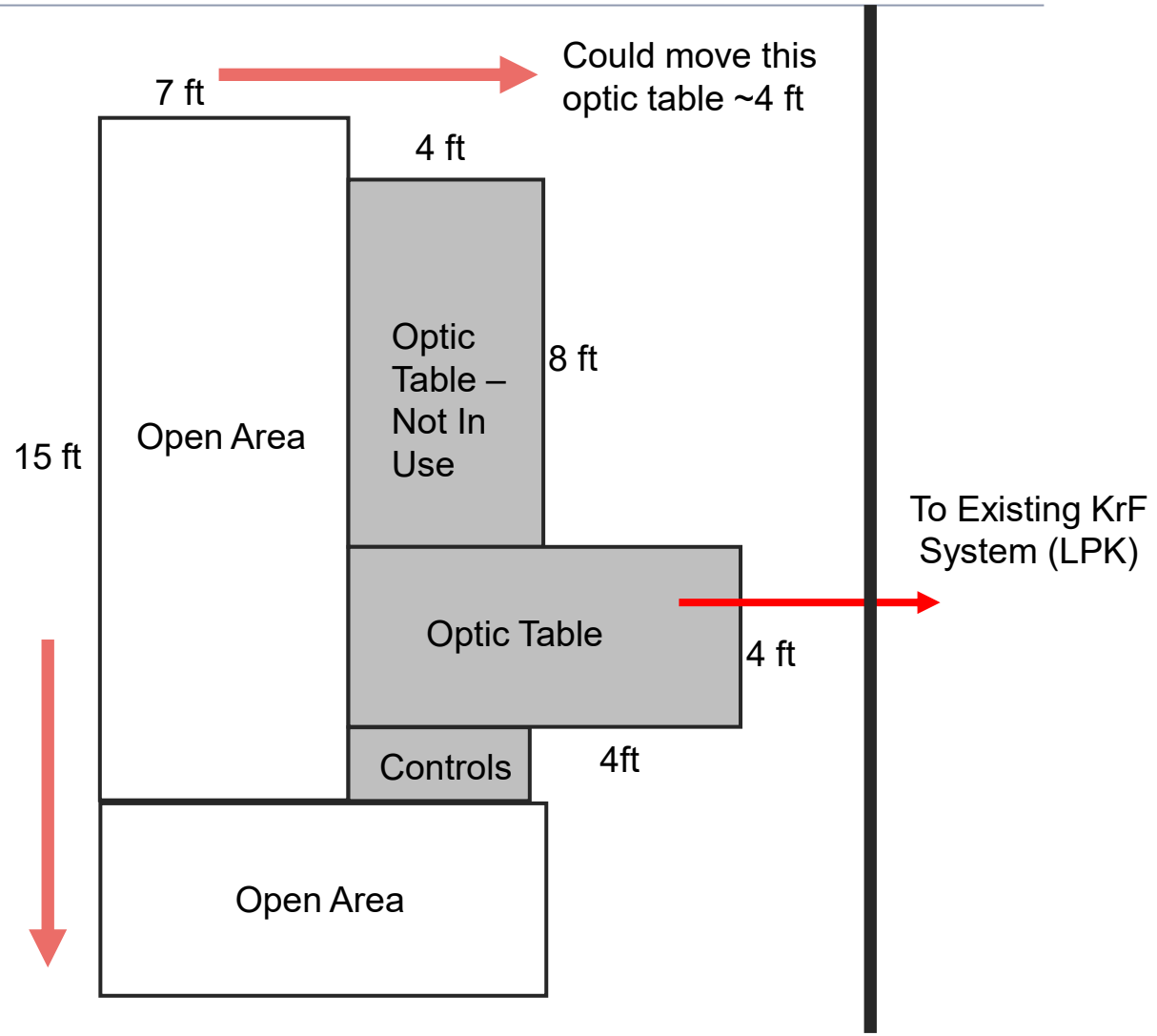
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Quarter 1 Updates: M2.2 Design Experimental Setup – Facilities

Begin: 11/18/25 **End: 08/18/26**







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Quarter 1 Updates: M2.2 Design Experimental Setup – System Matrix

Begin: 11/18/25 **End: 08/18/26**

Sub-System	Short Descriptions	Questions	Utilities	Person(s)
Pulsed Power (Febetron)	Pulsed power delivery to cathode. Time, Current, Voltage, Cathode Dimensions, Source/Load Impedance	Variable pulse time and power?	-120 or 208 V -Current Draw?	Dmitrii, Summer, Jake, Kevin
Electron Beam/Diode	Electron beam delivery to gas/plasma/laser cell. Helmholtz magnet. Current Density, Cathode/Hibachi Dimensions and Composition, Acceleration Voltage, A-K gap.	Do we need magnetic field? What strength if so?	-Vacuum system -Magnet?	Dmitrii, Summer, Kevin
Microwave Delivery	Microwave delivery to gas/plasma/laser cell. Power, Frequency, Waveguide Design	Variable pulse time, frequency, and power?	-120 to start -208/480V later -Cooling Water	Jake, Dmitrii, Kevin
Diagnostics	S11/S21 for microwave power. OES. KrF Laser Gain. Gas Impurities?, Absorption, Laser Scattering	Counterpropagation of e-beam and microwave? Future TS studies	-120V	Kevin, Austin, Andrey
Controls	Gas Mixing/Delivery, Vacuum System for E-beam Diode, Timing System, Magnets?		-120V	Brian, Kevin
Optics	Cavity Mirrors?, OES Collection, Laser Absorption/Scattering		-120 V -Cooling water	Kevin, Austin, Andrey
Gas/Plasma/Laser Chamber	Gas-filled chamber containing lasing medium and interfaces for all sub-systems. Chamber Dimensions, Interfaces, faraday Cup for E-beam?	How big of a chamber? Gas mixing/delivery. Va cuum.	-Gas -Vacuum -Pressurized	All

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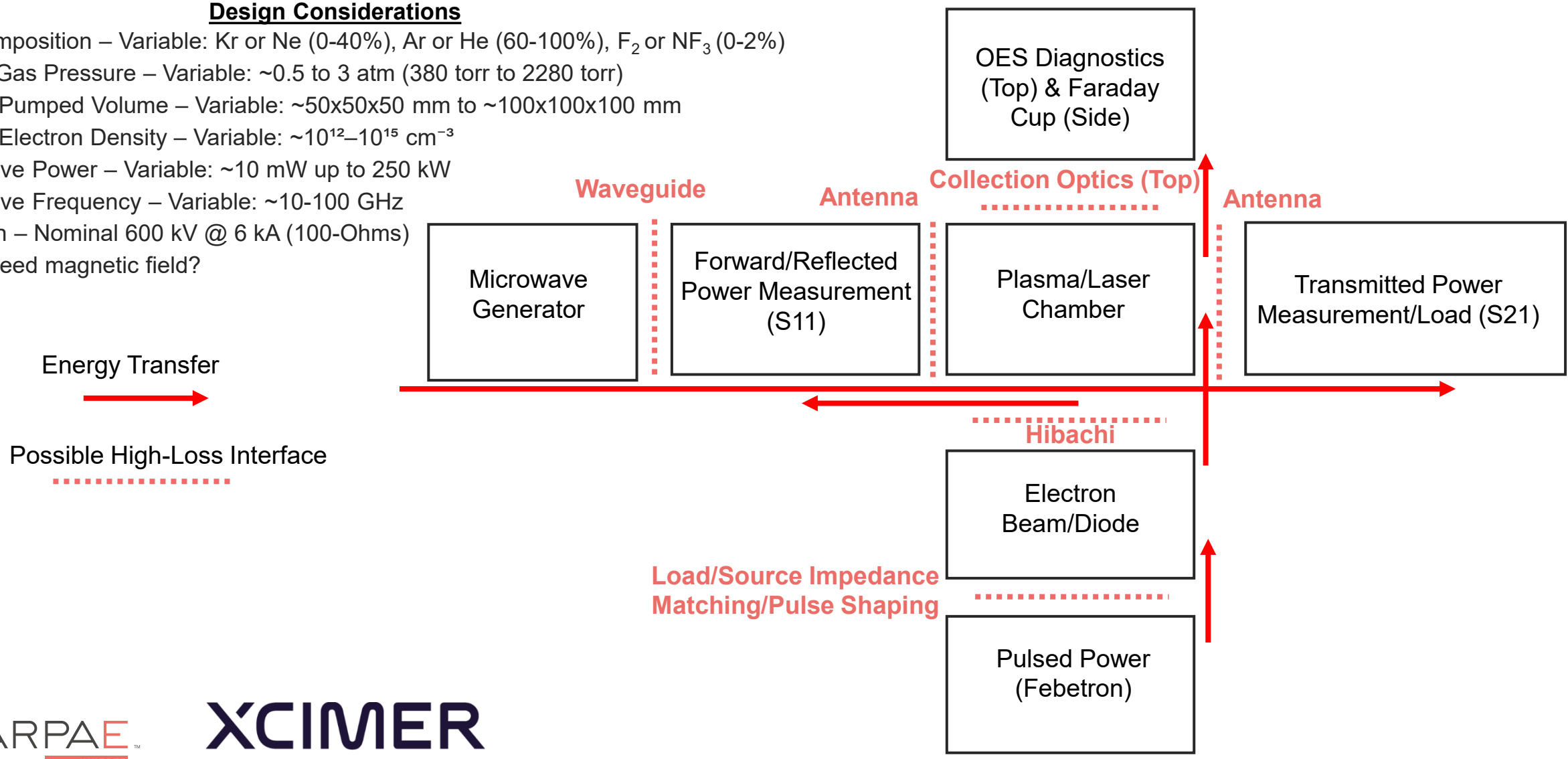
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Quarter 1 Updates: M2.2 Design Experimental Setup – Block Diagrams

Begin: 11/18/25 **End: 08/18/26**

Design Considerations

- Gas Composition – Variable: Kr or Ne (0-40%), Ar or He (60-100%), F_2 or NF_3 (0-2%)
- Overall Gas Pressure – Variable: ~0.5 to 3 atm (380 torr to 2280 torr)
- Plasma Pumped Volume – Variable: ~50x50x50 mm to ~100x100x100 mm
- Plasma Electron Density – Variable: ~10^{12}–10^{15} cm^{-3}
- Microwave Power – Variable: ~10 mW up to 250 kW
- Microwave Frequency – Variable: ~10-100 GHz
- Febetron – Nominal 600 kV @ 6 kA (100-Ohms)
- Do we need magnetic field?



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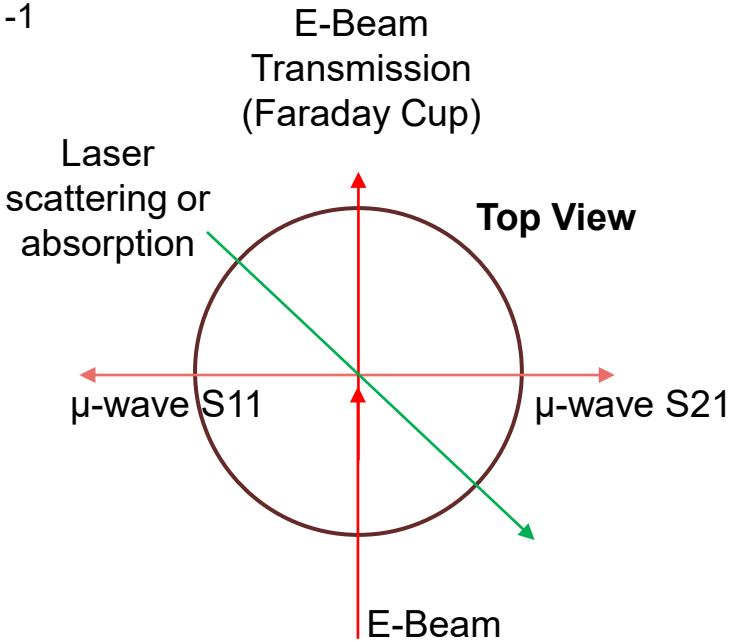
Quarter 1 Updates: M2.2 Design Experimental Setup – Block Diagrams

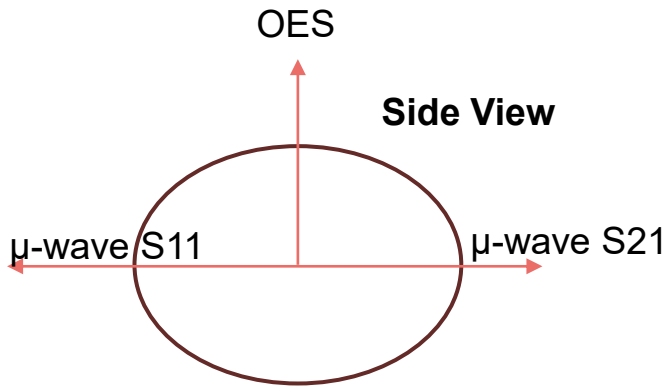
Begin: 11/18/25 **End: 08/18/26**

of Ports

1. Microwave S11 and S21 – 2
2. E-beam Delivery – 1
3. E-beam Transmission (Faraday Cup) – 1 Opposite of E-beam delivery side (2 total)
4. Plasma Diagnostics OES – 1 (out top)
5. Future Plasma Diagnostics (Laser Scattering/Absorption) – 2 (@ 135 degrees backscatter angle)
6. Gas Delivery – 1
7. Pressure Gauge – 1
8. Vacuum -1

-7 total ports for all experimental measurements
-3 ports for auxiliary controls
-3 extra ports (future)





An Option: Kimball Physics - 6” Vacuum Chamber
Quoted ~$13k



Spherical Square
UHV vacuum chamber
(2) 6.00"CF, (4) 4.50"CF, and (4) 1.33"CF
Grabber Grooves for internal mounts on 4.50" / 6.00" CF ports
Spherical ID: 5.90" (149.86 mm)
Internal volume: 119.0 cu. in. (1,950 cc)
Weight: 15.60 lbs (7.08 kg)

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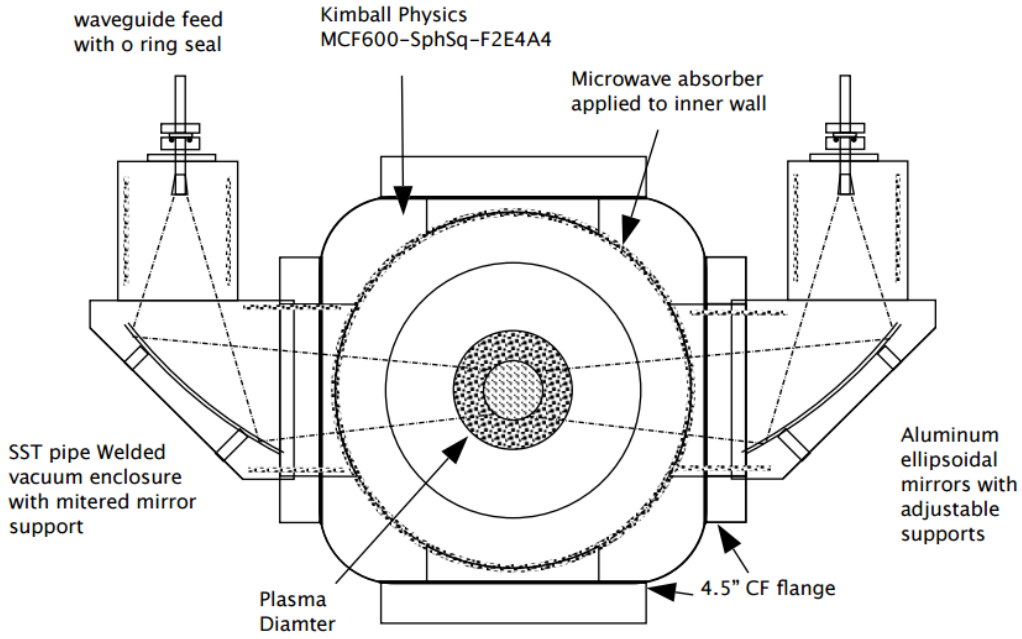
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Quarter 1 Updates: M2.2 Design Experimental Setup – Microwave Testing Schematics

Begin: 11/18/25 **End: 08/18/26**

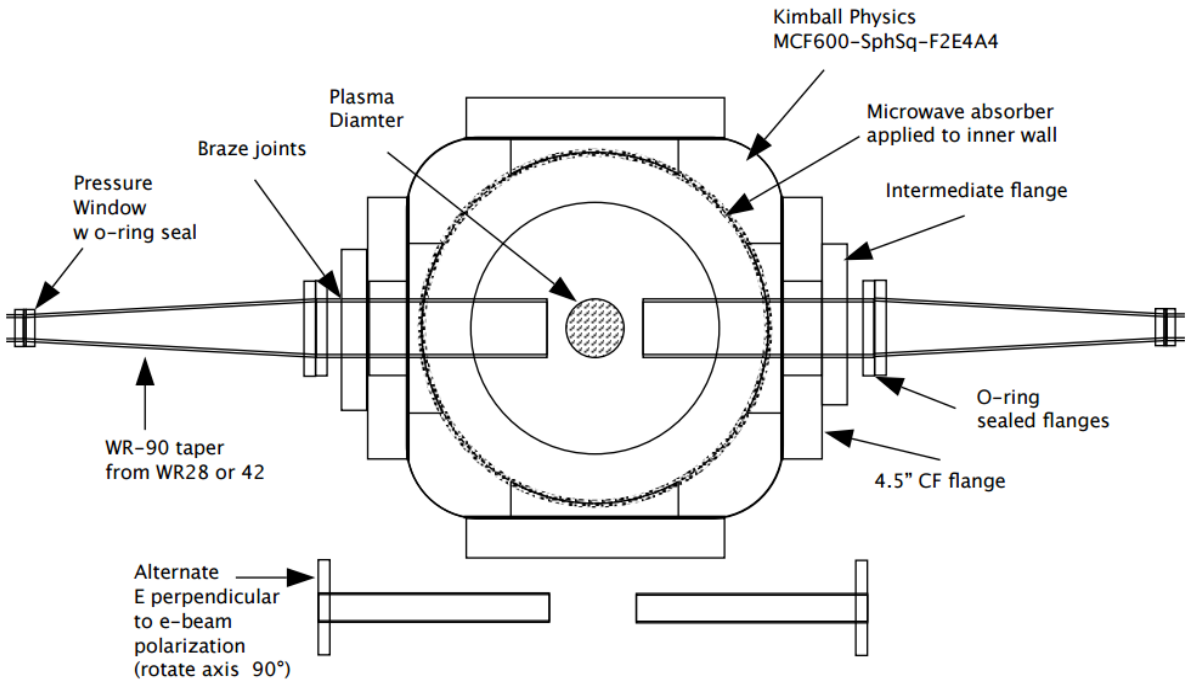
Free-Space Beamforming



Proposed Microwave reflection and loss measurement setup using ellipsoidal mirrors and small waveguide launchers

generated by MRP
Xcimer Proprietary

WR-90 Guided Wave



Proposed Microwave reflection and loss measurement setup using rectangular WR-90 waveguide launchers

generated by MRP
Xcimer Proprietary

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Quarter 1 Updates: M2.2 Design Experimental Setup – E-Beam Power Density (Kr/Ar Ionization and F_2 Attachment)

Begin: 11/18/25 **End: 08/18/26**

Electron Beam Power Density

$$P_{vol} = \frac{kW}{cm^{-3}}$$

Volumetric Ionization Rate

$$R_{ion} = \frac{P_{vol}}{W} \qquad W = \frac{J}{e^- - ion\ pair}$$

1J = 6.2415E18 eV

$$Ar \approx 26\ eV/e^- - ion\ pair$$
$$Kr \approx 24\ eV/e^- - ion\ pair$$

Dominant Electron Loss Mechanism

$$F_2 + e^- \rightarrow F + F^-$$

$$k_{attach} = 0.5 \text{ to } 2 \times 10^{-9}\ cm^3\ s^{-1}$$

$\uparrow T_e \quad \downarrow T_e$

Electron Density Primary Kinetics

$$\frac{dn_e}{dt} \approx R_{ion} - k_{attach}(n_e)[F_2]$$

$$\frac{dn_e}{dt} \approx \frac{P_{vol}}{W} - k_{attach}(n_e)[F_2]$$

$$P_{vol}\left(\frac{kW}{cm^{-3}}\right) = \frac{k_{attach}(n_e)[F_2] * W}{1000}$$

Estimated Electron Beam Power Density

$$n_e = 1 \times 10^{13}\ cm^{-3}$$

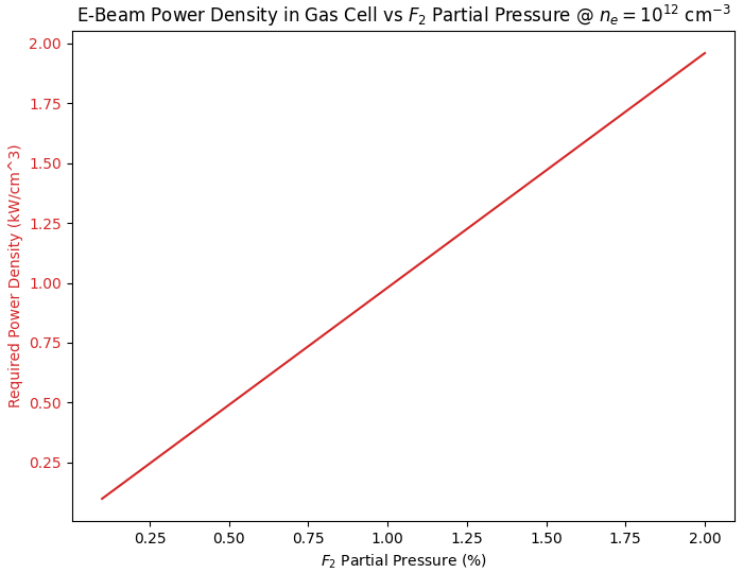
$$F_2 = 1\% = 2.45 \times 10^{17}\ cm^{-3} \text{ @ 1 atm and 300 K}$$

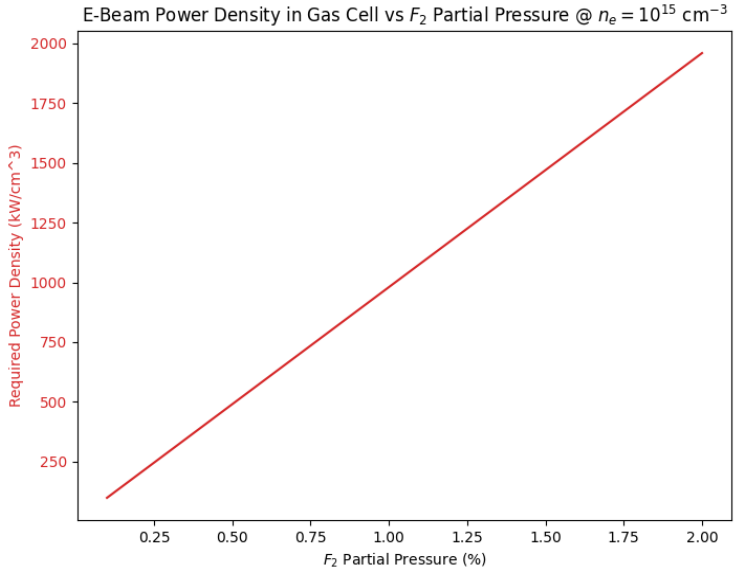
$$k_{attach} = 1 \times 10^{-9}\ cm^3\ s^{-1}$$

$$W = 25\ eV = 4.005 \times 10^{-18}\ J$$

$$P_{vol}\left(\frac{kW}{cm^{-3}}\right) = \frac{1 \times 10^{-9} \times (1 \times 10^{13}) \times 2.45 \times 10^{17} \times 4.005 \times 10^{-18}\ J}{1000}$$

$$\boldsymbol{P_{vol} = 9.8\left(\frac{kW}{cm^{-3}}\right)}$$





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Quarter 1 Updates: M2.2 Design Experimental Setup – E-Beam Current Density (Kr/Ar Ionization and F_2 Attachment)

Begin: 11/18/25 **End: 08/18/26**

Estimated Electron Beam Power Density (Low)

$$n_e = 1 \times 10^{12}\ cm^{-3}$$

$$F_2 = 0.1\% = 2.45 \times 10^{16}\ cm^{-3} \text{ @ 1 atm and 300 K}$$

$$P_{vol}\left(\frac{kW}{cm^{-3}}\right) = \frac{1 \times 10^{-9} \times (1 \times 10^{12}) \times 2.45 \times 10^{16} \times 4.005 \times 10^{-18}\ J}{1000}$$

$$\boldsymbol{P_{vol} = 0.098\ \left(\frac{kW}{cm^{-3}}\right)}$$

Estimated Electron Beam Power Density (High)

$$n_e = 1 \times 10^{15}\ cm^{-3}$$

$$F_2 = 2\% = 4.89 \times 10^{17}\ cm^{-3} \text{ @ 1 atm and 300 K}$$

$$P_{vol}\left(\frac{kW}{cm^{-3}}\right) = \frac{1 \times 10^{-9} \times (1 \times 10^{15}) \times 4.89 \times 10^{17} \times 4.005 \times 10^{-18}\ J}{1000}$$

$$\boldsymbol{P_{vol} = 1958.45\ \left(\frac{kW}{cm^{-3}}\right)}$$

Electron Beam Current Density

$$J = \frac{P_{vol} \times L}{V}$$

L = Stopping Length
V = Acceleration Voltage

Estimated Electron Beam Current Density

L < 5 cm @ 100 GHz (abs. = 0.2/cm)

$$J\left(\frac{A}{cm^2}\right) = \frac{0.098 \times 10^3 \left(\frac{W}{cm^{-3}}\right) \times 5\ cm}{600 \times 10^3 (V)}$$

$$\boldsymbol{J = 0.000817\ \left(\frac{A}{cm^2}\right)}$$

$$J\left(\frac{A}{cm^2}\right) = \frac{1958.45 \times 10^3 \left(\frac{W}{cm^{-3}}\right) \times 5\ cm}{600 \times 10^3 (V)}$$

$$\boldsymbol{J = 16.320\ \left(\frac{A}{cm^2}\right)}$$

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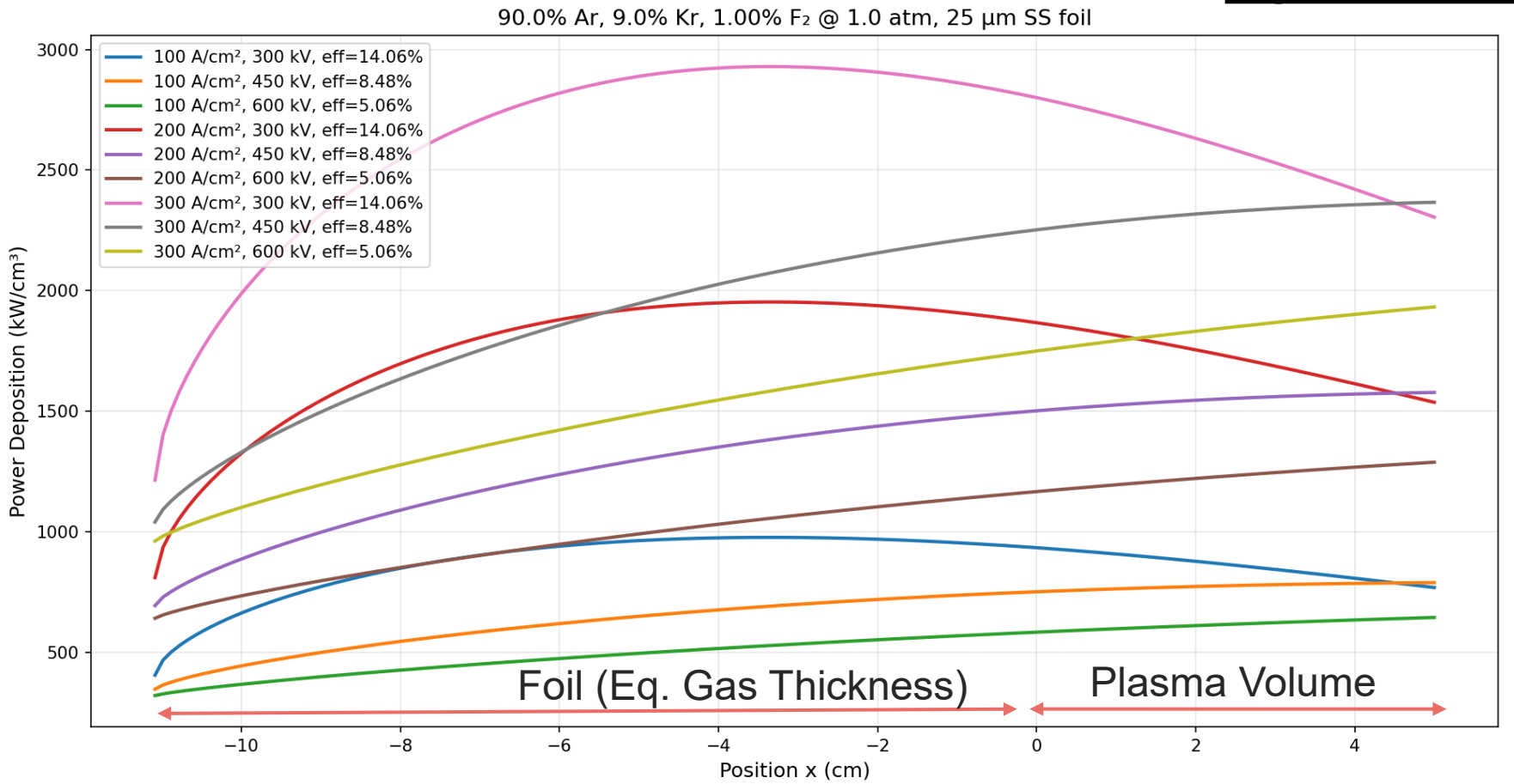
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Quarter 1 Updates: M2.2 Design Experimental Setup – Simulated E-Beam Deposition (Foil/Gas Losses)

Begin: 11/18/25 **End: 08/18/26**

To Sustain $n_e = 10^{15}$ cm^{-3} in a 5 cm plasma diameter
Critical power densities in plasma ≈ 1,960 kW cm^{-3}

High Plasma Electron Density



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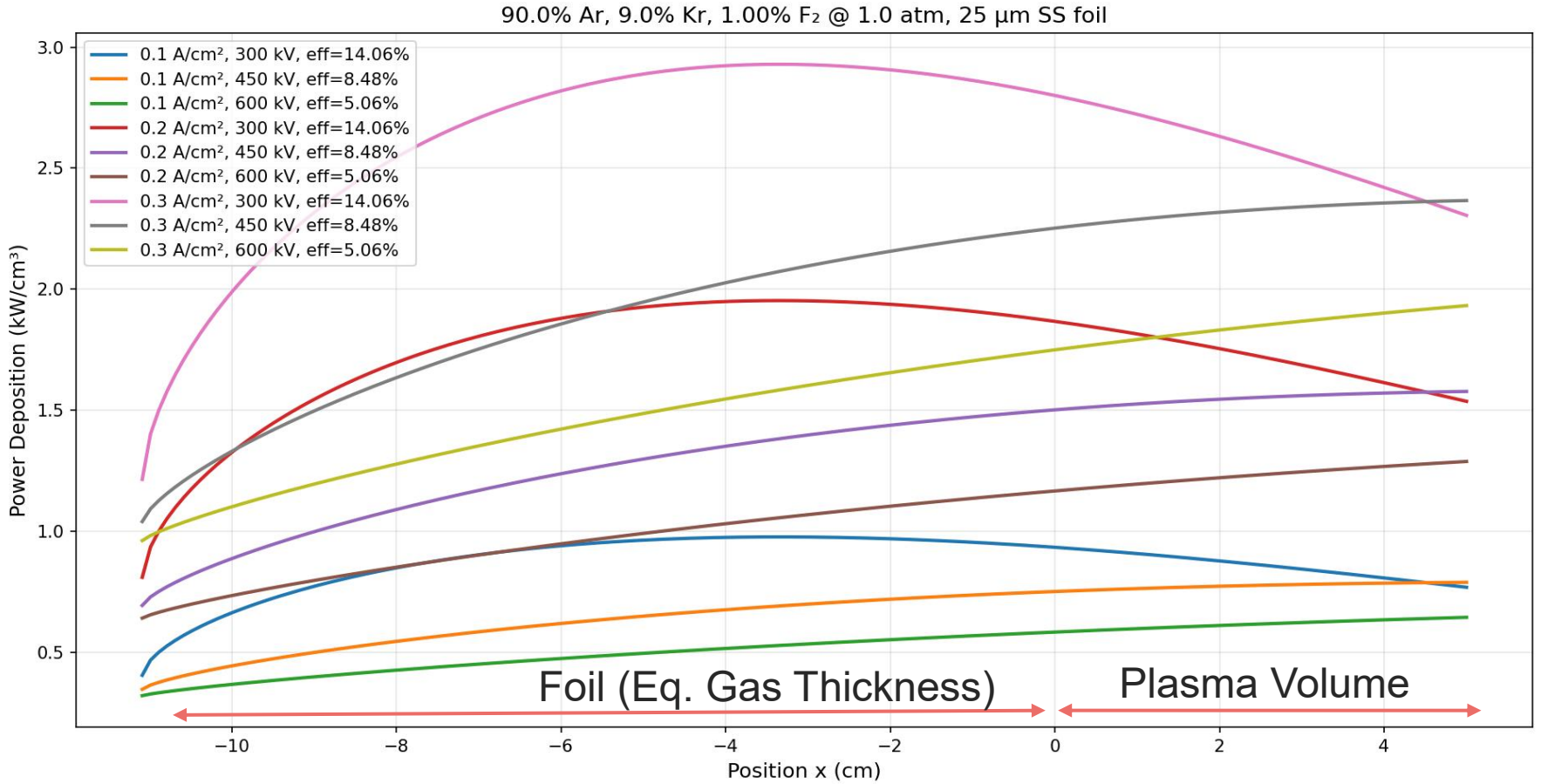
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Quarter 1 Updates: M2.2 Design Experimental Setup – Simulated E-Beam Deposition (Foil/Gas Losses)

Begin: 11/18/25 **End: 08/18/26**

To Sustain $n_e = 10^{12}$ cm^{-3} in a 5 cm plasma diameter
Critical power densities in plasma ≈ 0.098 kW cm^{-3}

Low Plasma Electron Density

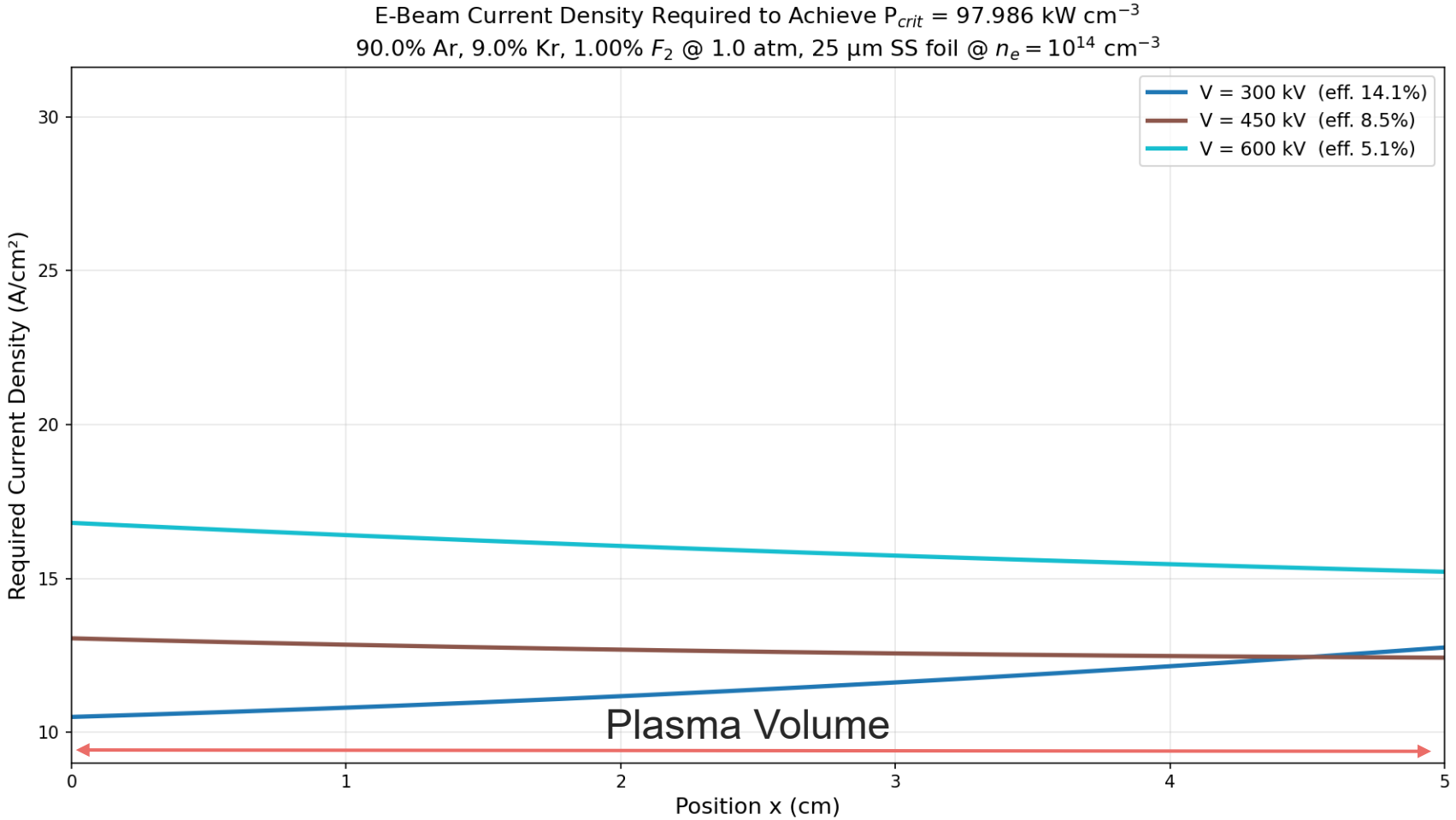


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Quarter 1 Updates: M2.2 Design Experimental Setup – Critical E-Beam Current Density (Typical Conditions)

Begin: 11/18/25 **End: 08/18/26**

To Sustain $n_e = 10^{14}$ cm^{-3} in 5 cm plasma diameter under typical KrF laser conditions



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Project Expenditure

- Q1 Spending under budget due to building team
- Q2+ Spending will significantly increase

		Total Project Cost			11/18/2025-2/16/2026			Cumulative Expenditure to Date			Remaining Balance
	CATEGORY	Total Project Cost	ARPA-E	Cost Share	Total Expense	Federal Reimbursement Request	Cost Share Amount	Cumulative Charges To-Date	ARPA-E	Cost Share	ARPA-E
a.	a. Personnel	$1,080,295.00	$1,080,295.00	$ -	$ 592.08	592.08	-	$ 592.08	$ 592.08	$ -	$1,079,702.92
b.	b. Fringe Benefits (39.95%)	$ 431,578.00	$ 431,578.00	$ -	$ 236.54	236.54	-	$ 236.54	$ 236.54	$ -	$ 431,341.46
c.	c. Travel	$ 22,290.00	$ 22,290.00	$ -	$ -	-	-	$ -	$ -	$ -	$ 22,290.00
d.	d. Equipment	$ 976,887.00	$ 237,187.00	$ 739,700.00	$ 98,890.00	-	98,890.00	$ 98,890.00	$ -	$ 98,890.00	$ 237,187.00
e.	e. Supplies	$ 229,619.00	$ 211,049.00	$ 18,570.00	$ -	-	-	$ -	$ -	$ -	$ 211,049.00
f.	f. Contractual	$ -	$ -	$ -		-	-	$ -	$ -	$ -	$ -
	Sub-recipient	$1,199,150.00	$1,199,150.00	$ -	$ -	-	-	$ -	$ -	$ -	$1,199,150.00
	FFRDC	$ -	$ -	$ -	$ -	-	-	$ -	$ -	$ -	$ -
	Vendor	$ 415,500.00	$ 220,500.00	$ 195,000.00	$ -	-	-	$ -	$ -	$ -	$ 220,500.00
	Total Contractual	$1,614,650.00	$1,419,650.00	$ 195,000.00	$ -	-	-	$ -	$ -	$ -	$1,419,650.00
g.	g. Construction	$ -	$ -	$ -	$ -	-	-	$ -	$ -	$ -	$ -
h.	h. Other Direct Costs	$ 169,200.00	$ 169,200.00	$ -	$ 525.00	525.00	-	$ 525.00	$ 525.00	$ -	$ 168,675.00
i.	i. Indirect Charges (15%)	$ 304,947.00	$ -	$ 304,947.00	$ 203.04	-	203.04	$ 203.04	$ -	$ 203.04	$ -
	Total Project Costs	$4,829,466.00	$3,571,249.00	$1,258,217.00	$100,446.66	1,353.62	99,093.04	$100,446.66	$ 1,353.62	$ 99,093.04	$3,569,895.38

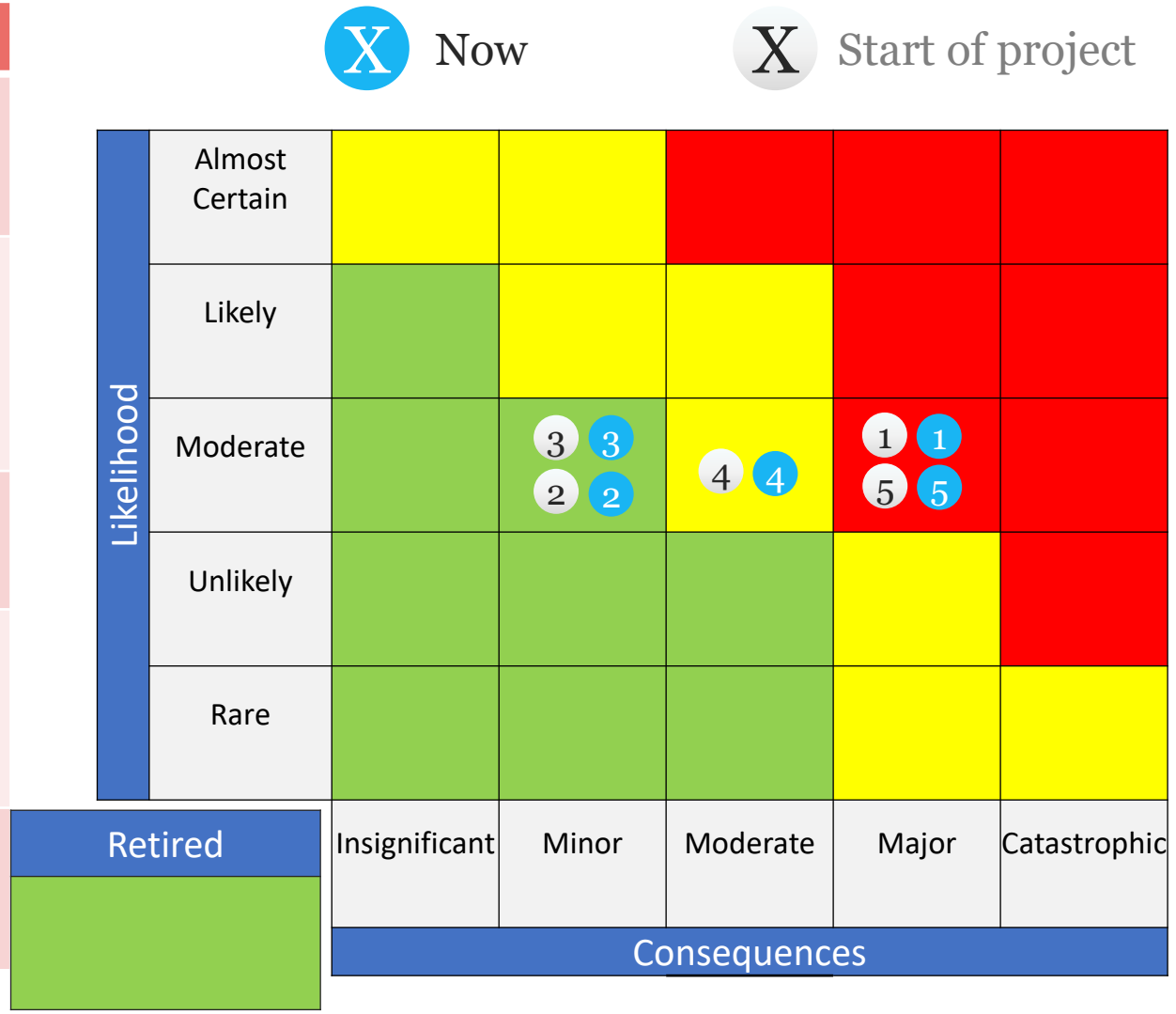
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Risk Matrix

#	Risk	Mitigation
1	>100 kW pulsed microwave source is not available for lease, or the source does not meet needed specs.	Identify alternative vendors/sources.
2	The plasmas under study exhibit strong absorption of 100 GHz radiation, preventing effective penetration into the plasma volume.	a) Select an alternative microwave source operating at a higher frequency (>100 GHz). b) Use plasmas with lower electron number density, potentially requiring a different gas composition.
3	Electron beam does not produce plasmas with specified characteristics.	Redesign and rebuild the electron gun, and the electron transmission window for improving beam delivery.
4	The fluorescence efficiency of the hybrid-pumped system shows no improvement or is even suppressed due to unexpected kinetic effects.	Use alternative gas mixtures containing different fluorine donors or other gases that help mitigate undesirable kinetic effects.
5	Microwave sources with the power levels required to pump laser systems are expensive to produce and/or impractical to implement.	Invest in the development of new, more economically viable microwave source designs (note: this is outside the scope of the current program).



ARPAE AWARDEE

XCIMER ENERGY CORPORATION

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Invention and Intellectual Property – Quarter 1 Updates

List of relevant pre-award IPs (Background IP)

- Pre-existing laser system concepts and architectures developed prior to Award Work.
- Internal modeling and analysis tools, including software or source code, used for laser system design/implementation developed prior to Award Work.
- General engineering methods and know-how related to large-scale laser system design/implementation developed prior to Award Work.
- Non-public technical data generated at Xcimer prior to Award Work.
- Existing patent filings and applications held by Xcimer as of the effective date of the Award.

List of new inventions made as a result of the award

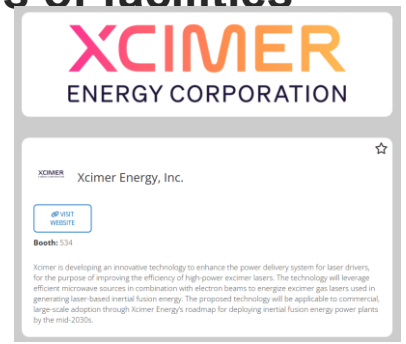
- None to date

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T2M Plan and Updates

- **Xcimer Energy, Inc was founded in 2022 and has secured over $100M in private capital funding**
  - Investors: Breakthrough Energy Ventures, Prelude Ventures, Lowercarbon Capital, Emerson Collective, and Hedosophia Investments.

- **Xcimer's roadmap to a commercial powerplant involves 3-generations of facilities**
  - Phoenix
  - Vulcan
  - Athena

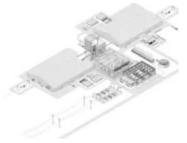


- **Registered to attend ARPA-E Energy Summit**
  - Dr. Kevin Finch and Dr. Andrey Mironov

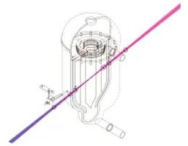
Facility and Major Components	Completion Date	Purpose
Phoenix • Long-pulse Kinetics Platform (LPK) • SBS Pulse Compression	March 2026	Demonstrate Xcimer laser architecture. Optimize excimer kinetics.
Vulcan • Argos excimer Amplifier Module (~200 kJ) • Full Beamline, 4 MJ on-target	Q4 2030	Full-scale laser prototype, delivering 4 MJ to target, achieving wall-plug breakeven.
Athena • Rep-rated Argos Modules • HYLIFE Chamber • 150 MW electric commercial plant	Mid-2030s	Commercial power plant delivering energy to the grid.



Accelerating the shift to abundant energy. Xcimer is building the world's biggest, brightest and most powerful lasers to commercialize the only scientifically validated form of fusion energy on Earth.



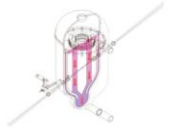
Xcimer combines novel laser technology with proven science to commercialize laser fusion energy. Founded in 2022 and based in Denver, Colorado, Xcimer is backed by leading climate-tech investors and supported by the U.S. Department of Energy. Our mission is to deliver clean, safe, reliable energy for the future.



10+ MJ Krypton Fluoride Laser - Our laser combines nonlinear "gas" optics with excimer laser amplifiers to produce very high beam energies at low cost. Excimer lasers have seen wide commercial application in semiconductor lithography, medical, and industrial applications.



Fusion Fuel Capsule - Like NIF, our fuel capsules are consumables containing hydrogen (DT) fuel that are ignited by the laser, but at much larger size and mass, resulting in higher performance, easier manufacturing, and more robust operation. The higher performance means a power plant can operate at lower repetition rates (<1 Hz) than conventional IFE concepts.



HYLIFE III FLiBe Chamber - Pioneered at Livermore National Lab and UC Berkeley over several decades, the chamber uses a self-contained liquid flow of lithium salt, or FLiBe, to absorb fusion output and completely protect the first structural wall.

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Next Steps & Work In Progress

- **Finalize IP/Data Management Plan**
  - Upload to ePIC system
- **Finalize subaward contracts with TAMU and UIUC**
- **Finish initial system design**
  - Plumbing and Instrumentation Diagram (P&ID)
  - Concept of Operations (ConOPs)
  - Architecture description design (ADD)
  - Most important requirements (MIPs)
- **Purchase all long (>3-4 months) lead-time equipment**
- **Verify/Install facility needs and initial safety/access workplan**
  - Water
  - Power
  - Exhaust
  - Radiation/EMI Shielding
  - Access controls

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